Automated measure of fetal head circumference

Truong Thi Lan Anh
University of Science and Technology of Hanoi
Ha Noi, Vietnam

I. Introduction

Accurate measurement of fetal head circumference (HC) from ultrasound images is an important task in prenatal care because it helps doctors evaluate fetal growth and detect possible abnormalities. However, manual measurement is time-consuming and may vary between operators due to differences in experience and image quality. Therefore, automated methods using machine learning and deep learning are widely studied to support reliable and fast measurement.

In this project, we use the HC18 ultrasound dataset, which provides fetal head ultrasound images along with pixel size information and head circumference values for the training set. The goal is to build a deep learning model that can predict head circumference (in millimeters) from an input ultrasound image. We implement a model and adapt it for regression by extracting deep features from the network and estimating the HC value. Model performance is evaluated using Mean Absolute Error (MAE), which measures the average absolute difference between predicted and true head circumference values. This evaluation metric is appropriate for regression tasks and provides an interpretable error in millimeters.

II. Dataset

We use the HC18 fetal head ultrasound dataset. It contains grayscale 2D ultrasound images and a CSV file that provides the image filename, pixel size (mm/pixel), and (for the training set) the head circumference (HC) label in millimeters.

a) Files and size: The dataset is split into:

- `training_set`: 999 ultrasound images with labels in `training_set_pixel_size_and_HC.csv`
- `test_set`: 335 ultrasound images with pixel size in `test_set_pixel_size.csv` (no HC labels)

b) Sample images: Figure 1 shows random examples from the training set.

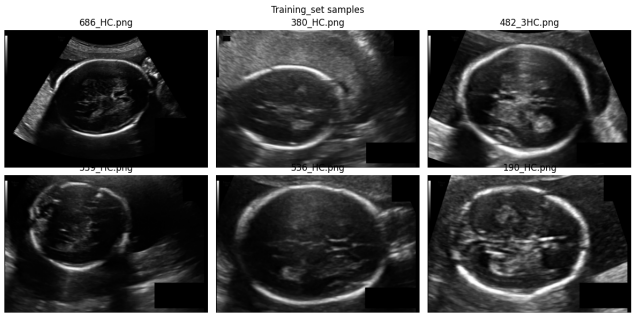

Fig. 1. Example ultrasound images from the training set.

c) Training statistics: Table I summarizes the training CSV statistics used in this work.

TABLE I
Summary statistics from `training_set_pixel_size_and_HC.csv` (999 samples).

Statistic	Pixel size (mm/pixel)	HC (mm)
Count	999	999
Mean	0.139846	174.383093
Std	0.053005	65.282061
Min	0.049415	44.300000
25%	0.108801	153.600000
50%	0.128357	174.060000
75%	0.157678	189.825000
Max	0.393280	346.400000

III. Methodology

A. Data preprocessing

All ultrasound images are loaded from `training_set`. Annotation images (filenames containing "Annotation") are ignored. Each image is converted to grayscale and resized to a fixed input size of 224×224, which matches the standard input size for EfficientNet-B0. Because EfficientNet expects 3-channel input, the grayscale image is duplicated across channels to form an RGB-like tensor. Finally, we apply the ImageNet normalization used by pretrained EfficientNet weights.

To evaluate performance, the training set is split into training and validation subsets 80/20. The official test set has no HC labels, so MAE is computed on the validation split.

B. Model architecture

We use EfficientNet-B0 as the backbone network with pretrained ImageNet weights. EfficientNet uses compound scaling to balance network depth, width, and input resolution, and it provides strong feature extraction performance with relatively low computation.

The original EfficientNet-B0 ends with a classification head. For regression, we replace the final classifier layer with a single-neuron linear layer:

$$\hat{y} = f_\theta(x),$$

where x is the input image and $\hat{y}$ is the predicted head circumference in mm. In practice, the final layer is changed from a fully connected layer with many classes to a linear layer with 1 output.

C. Training procedure

The model is trained end-to-end on the training split. We optimize using the L1 loss (Mean Absolute Error loss) between the prediction $\hat{y}$ and the ground truth y:

$$\mathcal{L}_{L1} = \frac{1}{N} \sum_{i=1}^{N} |\hat{y}_i - y_i| \,.$$

Adam is used as the optimizer with a small learning rate = 10^{-4}. After each epoch, we compute MAE on the validation set. The checkpoint with the lowest validation MAE is saved as the best model.

D. Evaluation metric

Model performance is measured using Mean Absolute Error (MAE) in millimeters:

$$MAE = \frac{1}{N} \sum_{i=1}^{N} |\hat{y}_i - y_i| \,.$$

MAE is suitable for regression and is easy to interpret because it directly indicates the average prediction error in mm.

IV. Results

We trained EfficientNet-B0 for 10 epochs and evaluated performance using Mean Absolute Error (MAE) in millimeters. Both training and validation MAE decreased over epochs, showing that the model learned useful features from the ultrasound images. The best performance was obtained at epoch 10.

Epoch	Train MAE (mm)	Val MAE (mm)
1	102.454	98.043
2	98.816	92.514
3	95.660	90.783
4	92.117	90.709
5	89.008	89.634
6	85.333	76.982
7	82.011	73.705
8	78.431	72.242
9	75.014	69.270
10	71.507	61.288

TABLE II
TRAINING AND VALIDATION MAE (MM) ACROSS EPOCHS FOR EFFICIENTNET-B0.

The lowest validation MAE achieved was **61.288 mm**, and the corresponding model checkpoint (`best_efficientnet_hc18.pt`) was saved for inference.

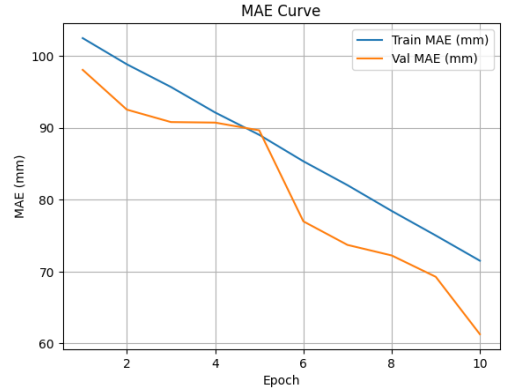

Fig. 2. Training and validation MAE curves across 10 epochs.

The MAE curve shows that the model improves steadily during training. Training MAE decreases from about 102 mm at epoch 1 to about 72 mm at epoch 10, while validation MAE decreases from about 98 mm to about 61 mm. A noticeable drop in validation MAE occurs after around epoch 6, and the best performance is achieved at the final epoch, indicating better generalization over time with no clear overfitting in this run.

V. References

Thomas L. A. van den Heuvel, Dagmar de Bruijn, Chris L. de Korte and Bram van Ginneken. Automated measurement of fetal head circumference using 2D ultrasound images. PloS one, 13.8 (2018): e0200412.

Ramesh, J., Bacher, V., Eid, M. C., Kalabizadeh, H., Rupprecht, C., Namburete, A. I. L., Yeung, P.-H., Wyburd, M. K., Dinsdale, N. K. (2025). *Automated Fetal Biometry Assessment with Deep Ensembles using Sparse-Sampling of 2D Intrapartum Ultrasound Images*. arXiv preprint arXiv:2505.14572.